

Experiment No: 07

Experiment Name: Design of a Fuzzy Logic Controller for Speed Control of a DC Motor.

Objectives:

1. To understand the principles of fuzzy logic and its application in real-world control systems.
2. To design a fuzzy inference system (FIS) for speed control of a DC motor using MATLAB.
3. To simulate the fuzzy logic controller and evaluate its performance under various input conditions.

Theory:

Fuzzy Logic Control (FLC) is a control methodology that handles imprecision and uncertainty inherent in real-world systems. Unlike conventional controllers that require an exact mathematical model, FLC operates on qualitative rules and approximate reasoning derived from human expertise.

The key components of a fuzzy logic controller are:

- Fuzzification: Converts crisp input values into fuzzy linguistic values using membership functions.
- Rule Base: A set of IF-THEN rules that define the control action based on fuzzy inputs.
- Inference Engine: Evaluates the applicable rules and combines the results.
- Defuzzification: Converts the aggregated fuzzy output back into a crisp control value using methods such as centroid.

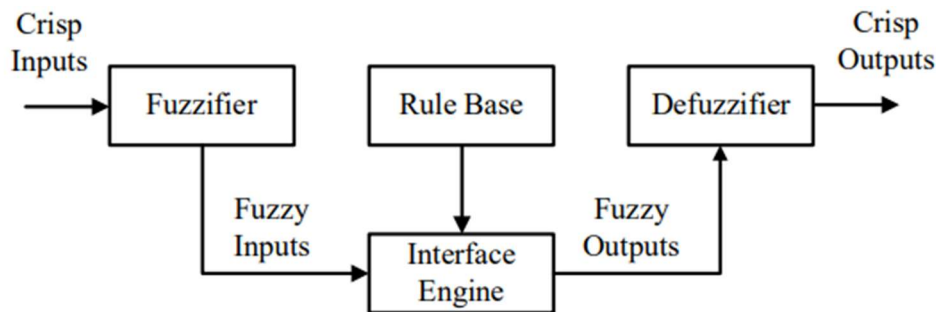


Figure 7.1: Block Diagram of Fuzzy Logic Controller

In this experiment, a Mamdani-type FLC is designed with two inputs — Error (e) and Derivative Error (de/dt) — and one output — Speed. Each variable is partitioned into five linguistic terms: Negative Big (NB), Negative Small (NS), Zero (ZE), Positive Small (PS), and Positive Big (PB).

Required Components:

1. Computer with MATLAB with Simulink
2. Fuzzy Logic Toolbox in MATLAB
3. Simulink environment for system simulation

System Model:

The DC motor system consists of an armature circuit with resistance R and inductance L , coupled to a mechanical rotor with moment of inertia J and friction coefficient b . The back-EMF constant and torque constant are both denoted by K . The motor's electrical and mechanical dynamics interact through the armature current and angular velocity.

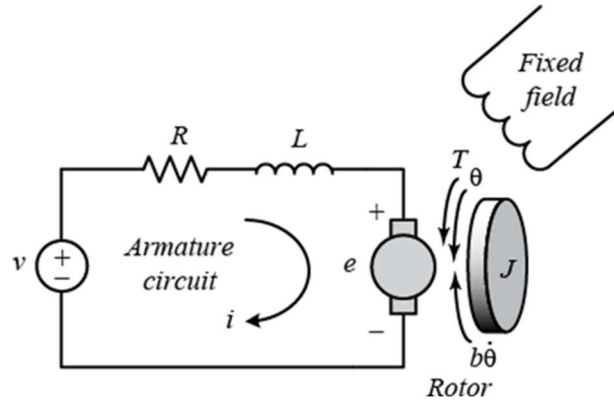


Figure 7.2: Representation of DC Motor.

Table 1: DC Motor Parameters

Parameter	Value
J ($\text{kg}\cdot\text{m}^2$)	0.01
b ($\text{N}\cdot\text{m}\cdot\text{s}$)	0.1
K ($\text{N}\cdot\text{m}/\text{A}$)	0.01
R (Ω)	1
L (H)	0.5

System Transfer Function:

The transfer function of the DC motor, relating angular displacement $\theta(s)$ to input voltage $V(s)$, is derived from the combined electrical and mechanical equations:

Motor system transfer function,

$$\frac{\dot{\theta}(s)}{V(s)} = \frac{K}{(Js + b)(Ls + R) + K^2}$$

Where,

- J = moment of inertia
- b = friction coefficient
- L = armature inductance
- R = armature resistance
- K = motor constant.

Simulink Model:

The closed-loop Simulink model consists of a step input, a summing junction to compute the error, a derivative block to calculate the error rate of change, the FLC block, a gain of 100, and the DC motor transfer function. The output is monitored via a Scope block.

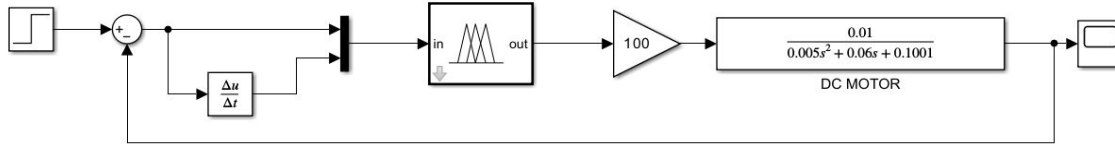


Figure 7.3: Simulink Model for FLC-Controlled DC Motor

Working Procedure:

1. Open MATLAB and launch the Fuzzy Logic Designer by typing fuzzyLogicDesigner in the Command Window.

The Fuzzy Logic Designer interface was opened, which allows defining the FIS structure, inputs, output, and inference method. The system was configured as a FIS named fuzzy1 with And method: min, Or method: max, Implication: min, Aggregation: max, and Defuzzification: centroid.

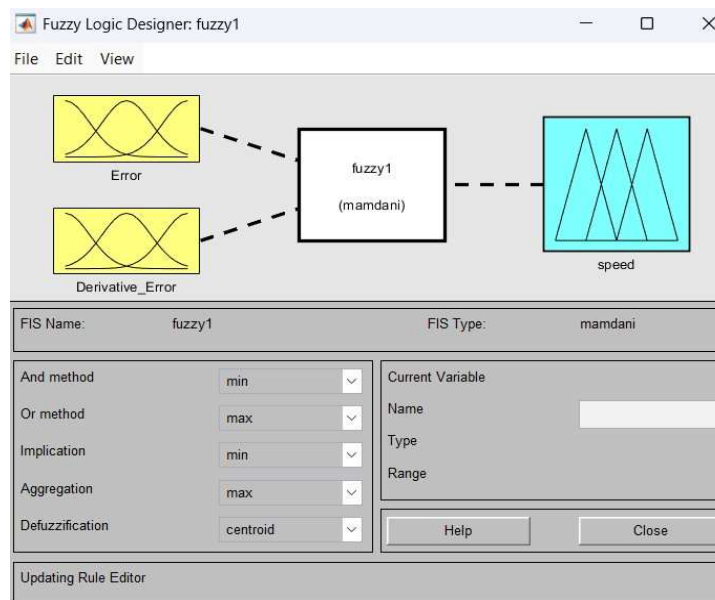


Figure 7.4: Fuzzy Logic Designer Interface

2. Define membership functions for the input variable Error over the range $[-1.5, 1.5]$.

Five triangular membership functions were defined for the Error input: NB $[-1.5 \ -1.5 \ -0.75]$, NS $[-1.5 \ -0.75 \ 0]$, ZE $[-0.75 \ 0 \ 0.75]$, PS $[0 \ 0.75 \ 1.5]$, and PB $[0.75 \ 1.5 \ 1.5]$.

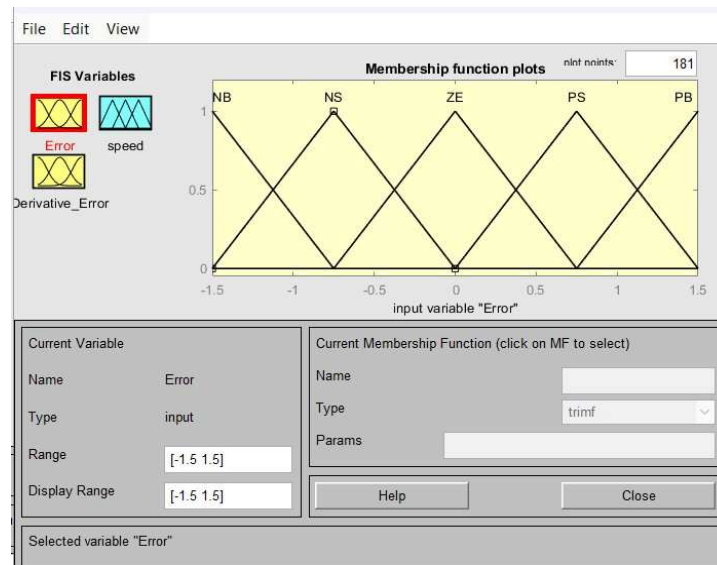


Figure 7.5: Membership Functions for Input Variable "Error"

3. Define membership functions for the input variable Derivative_Error over the range $[-1.5, 1.5]$.

The same five triangular membership functions were applied to the Derivative_Error input with identical parameters as the Error variable.

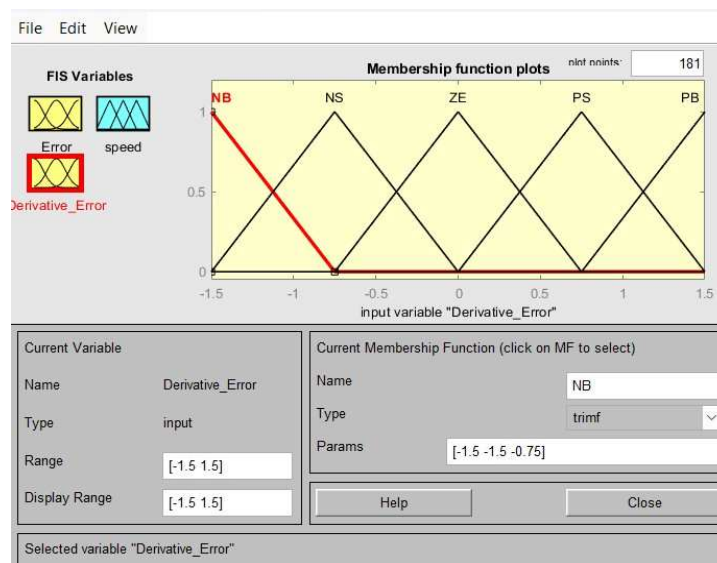


Figure 7.6: Membership Functions for Input Variable "Derivative_Error"

4. Define membership functions for the output variable Speed over the range $[-20, 20]$.

Five triangular membership functions were defined for the Speed output: NB $[-20 -20 -10]$, NS $[-20 -10 0]$, ZE $[-10 0 10]$, PS $[0 10 20]$, and PB $[10 20 20]$.

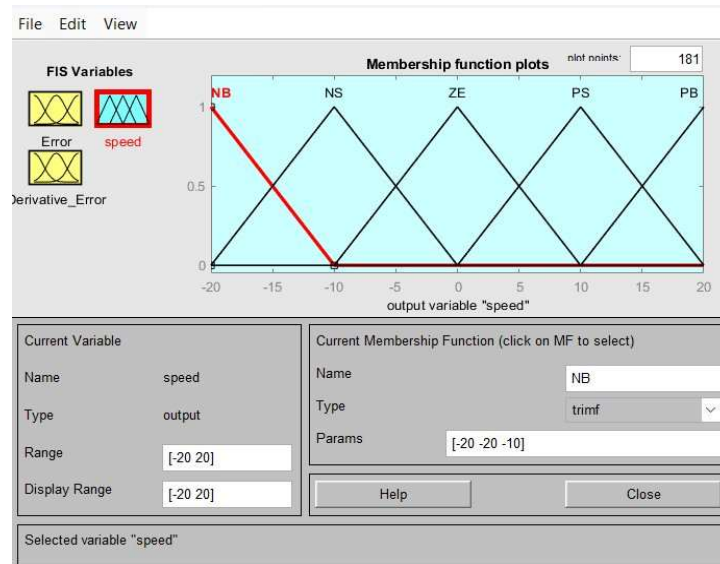


Figure 7.7: Membership Functions for Output Variable "Speed"

5. Construct the 5×5 fuzzy rule base using the Rule Editor.

A total of 25 IF-THEN rules were formulated based on the combination of Error and Derivative_Error linguistic values. The rule table is shown below:

Table 2: Fuzzy Rule Base (Output: Speed)

Error \ D_Error	NB	NS	ZE	PS	PB
NB	NB	NB	NB	NS	ZE
NS	NB	NB	NS	ZE	PS
ZE	NB	NS	ZE	PS	PB
PS	NS	ZE	PS	PB	PB
PB	ZE	PS	PB	PB	PB

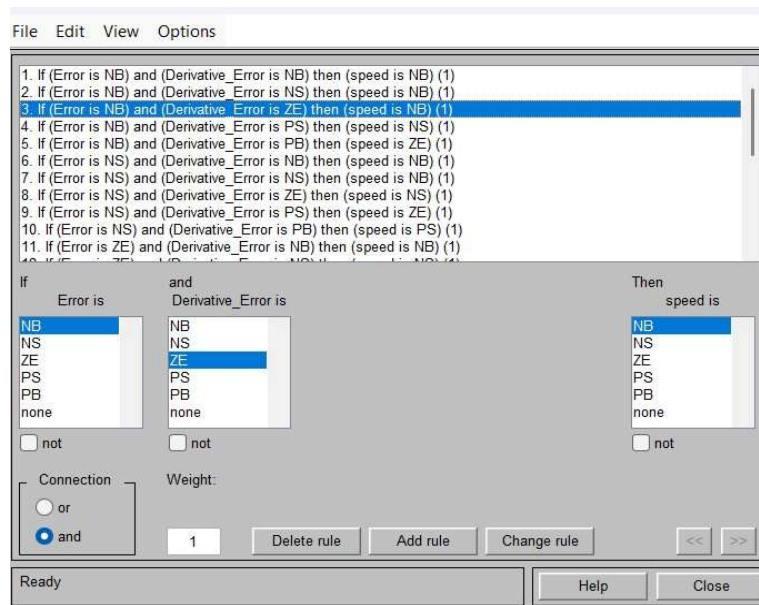


Figure 7.8: Rule Editor Showing All 25 Fuzzy Rules

6. Verify rule activation using the Rule Viewer for input [0; 0].

The Rule Viewer was opened to visualize the firing of all 25 rules for Error = 0 and Derivative_Error = 0. The resulting output speed was approximately -5.11×10^{-17} , effectively zero, confirming correct behavior at the operating point.

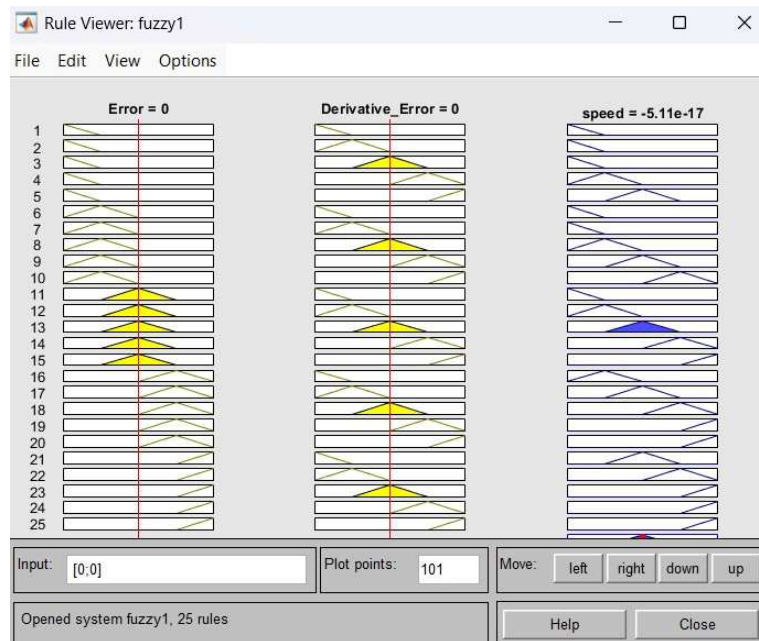


Figure 7.9: Rule Viewer — Activation of 25 Rules at Input [0, 0]

7. Examine the control surface using the Surface Viewer.

The Surface Viewer was used to display the 3D input-output mapping of the FLC. The surface shows a monotonically increasing relationship from negative speed (large negative error) to positive speed (large positive error), confirming the designed control behavior.

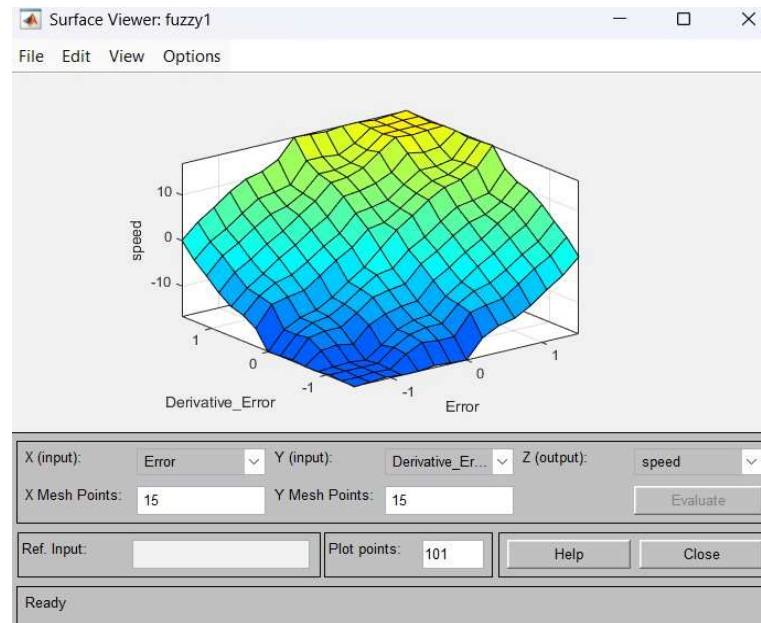


Figure 7.10: Control Surface — Speed as a Function of Error and Derivative_Error

8. Integrate the FIS into Simulink and run the simulation.

The saved FIS file (fuzzy1) was loaded into a Fuzzy Logic Controller block in the Simulink model. The model was simulated with a unit step input over a 10-second period, and the output was observed on the Scope.

Results:

The Simulink simulation was executed with a unit step reference input. The scope output is shown below:

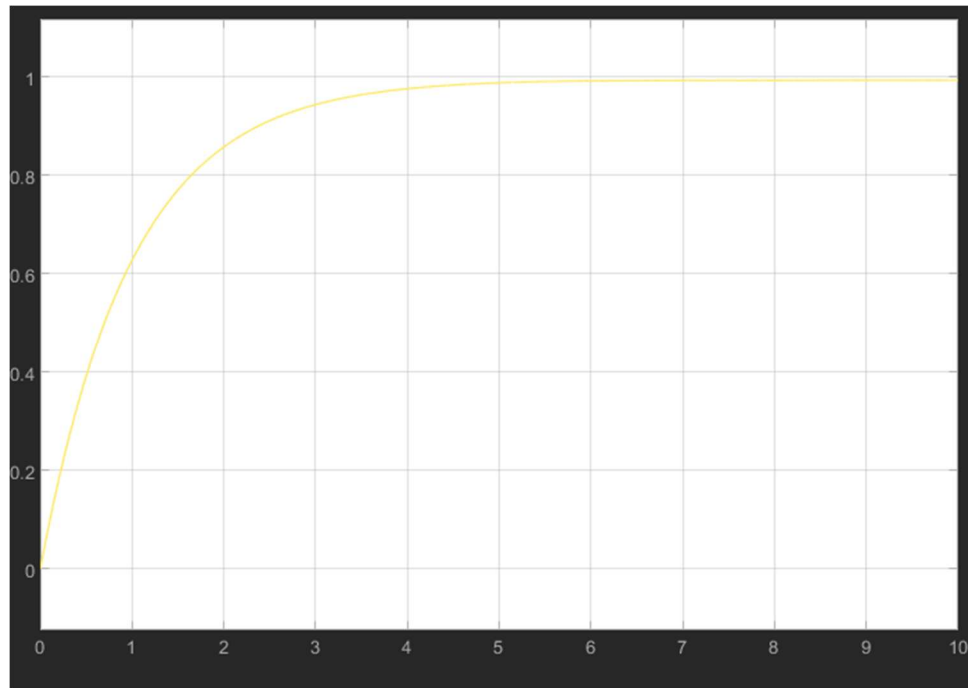


Figure 7.11: Simulink Scope Output — Step Response of FLC-Controlled DC Motor

The output response exhibits a smooth, overdamped step response with no overshoot. The system's speed output tracks the reference signal, settling to the desired value of 1.0 at approximately $t = 5\text{--}6$ seconds, demonstrating the effectiveness of the fuzzy logic controller in providing stable and accurate speed control.

Discussion:

The Mamdani fuzzy logic controller successfully controlled the DC motor speed without any steady-state error or overshoot. The use of 25 rules derived from a 5×5 rule table provided sufficient coverage of the input space. The centroid defuzzification method yielded smooth and physically meaningful control actions. The control surface confirmed anti-symmetric behavior, which is appropriate for a speed regulation task. Compared to a conventional PID controller, the FLC does not require an explicit mathematical model of the plant and is inherently robust to parameter variations.

Conclusion:

A Mamdani-type fuzzy logic controller was successfully designed and implemented in MATLAB/Simulink for DC motor speed control. The system utilized two inputs — Error and Derivative_Error — each with five triangular membership functions, and a corresponding output variable Speed over the range $[-20, 20]$. A 25-rule base was constructed and verified using the Rule Viewer and Surface Viewer. The closed-loop Simulink simulation confirmed that the FLC drives the motor speed to the desired setpoint smoothly with no overshoot, validating the fuzzy approach as an effective alternative to classical control strategies for nonlinear and uncertain systems.

References

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- [2] E. H. Mamdani and S. Assilian, "An Experiment in Linguistic Synthesis with a Fuzzy Logic Controller," *International Journal of Man-Machine Studies*, vol. 7, no. 1, pp. 1–13, 1975.
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